Department of Electrical & Computer Engineering
Cullen College of Engineering
University of Houston
ECE3441: Digital Logic Design
Spring 2016-Lab Exam

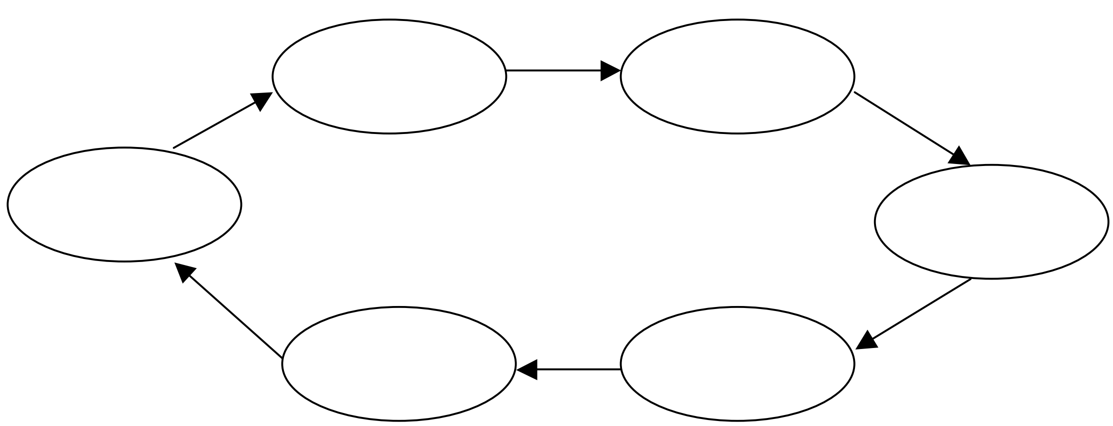
Student's Full Name: ____
Student ID: ____
Student signature: ____

Rules:

1. You have exactly one hour to complete the exam (including debugging). **NO EXTRA TIME** will be allowed.
2. You will be required to hand in **the details and the schematic** of your final design at the end of the exam.
3. You are allowed to use the TTL data book and a single-page (two-sided) cribsheet.

DESIGN PROBLEM:

1. Design and build a **minimum counter** circuit with the count sequence given in the NS-graph below.
2. Use **D flip-flops** for your design.
3. Make sure that unused states are designed to go to one of the states in the intended count sequence. Demonstrate this to the TA by forcing the counter (Use **PRE** and **CLR** inputs to do this) to start in each of the unused states.
4. Your report should consist of (1) the transition (NS) -table, (2) NS-maps, (3) K-maps, (4) circuit equations, (5) the circuit diagram, and (6) the details about debugging.



GRADING FOR THE LAB EXAM

Grade over 100

1) Incorrect design on paper…………………………………………………….…………..10
2) Correct design on paper but not optimal……………………………………...……..20
3) Correct design on paper, optimal solution…………………………………………….30
4) Correct design on paper but not optimal **and** circuit built, but circuit not working correctly…………………………………………………………………………………….35
5) Correct design on paper, optimal solution **and** circuit built, but circuit not working correctly……………………………………………………………………………….....40
6) Correct design on paper but not optimal **and** circuit built, circuit working correctly……………………………………………………………………………….....70
7) Correct design on paper, optimal solution **and** circuit built, circuit working correctly…………………………………………………………………………………….85

DEBUGGING: for 6) and 7) above, successful debugging gives **additional 15 points**.